BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Mathematics — SSC-I (9th Grade)

Syllabus: Chapter 3 — Sets and Relations

Total Marks: 65 **Time Allowed: 2h 30m** **Version: 1**

General Instructions:

1. Write your name, roll number, and section clearly on the answer sheet.
2. **Section A** (MCQs): Attempt **ALL 12** questions. Each carries **1 mark**. Time: 15 minutes.
3. **Section B** (Short Questions): Attempt **ALL 11** pairs, choosing the main question **OR** its alternative. Each carries **3 marks**.
4. **Section C** (Long Questions): Attempt **ALL 4** pairs, choosing the main question **OR** its alternative. Each carries **5 marks**.
5. Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

Encircle the correct option. All questions are compulsory.

#	Question	A	B	C	D	Answer
1	A set is a well-defined collection of:	similar objects	distinct objects	ordered objects	finite objects	○○○○
2	In set builder form, $A - B$ is written as:	$\{x \mid x \in A\}$	$\{x \mid x \in A \wedge x \notin B\}$	$\{x \mid x \in A \wedge x \in B\}$	$\{x \mid x \in B\}$	○○○○
3	If $A \cup B = A$ and $A \cap B = B$, then:	$B \subset A$	$A \subset B$	$A = B$	$A \neq B$	○○○○
4	Two sets A and B are disjoint if:	$A \cap B = A$	$A \cup B = \varnothing$	$A \cap B = \varnothing$	$A \subset B$	○○○○
5	Associative property of union states:	$(A \cup B) \cup C = A \cup (B \cap C)$	$(A \cup B) \cup C = A \cup (B \cup C)$	$A \cup (B \cup C) = (A \cap B) \cup C$	$A \cup B = B \cup A$	○○○○
6	Distributive property of union over intersection is:	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	$A \cup (B \cap C) = (A \cup B) \cup (A \cup C)$	$A \cap (B \cup C) = (A \cup B) \cap C$	$A \cup (B \cap C) = (A \cap B) \cup C$	○○○○
7	For overlapping sets A and B, $n(A \cup B)$ equals:	$n(A) + n(B)$	$n(A) \times n(B)$	$n(A) + n(B) - n(A \cap B)$	$n(A) - n(A \cap B)$	○○○○
8	If $n(A) = p$, then $n(A \times A)$ equals:	p	$2p$	p^2	$2p^2$	○○○○
9	Two ordered pairs (a, b) and (c, d) are equal if and only if:	$a = d$ and $b = c$	$a = c$ only	$a = c$ and $b = d$	$a + b = c + d$	○○○○
10	If $n(B) = t$, then number of binary relations in $B \times B$ is:	t	t^2	2^t	2^{t^2}	○○○○
11	Set of first elements of ordered pairs in a binary relation is called its:	Range	Domain	Co-domain	Cartesian product	○○○○
12	If $R = \{(2,1), (4,3), (2,2)\}$, then Dom R is:	$\{2, 4, 2\}$	$\{2, 4\}$	$\{1, 3, 2\}$	none of these	○○○○

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt ALL 11 questions. Choose either the main question OR its alternative.

Q.	Question	Marks	OR	OR Question	Marks
2(i)	Write the following set in set builder form: B = Set of prime numbers less than 17.	**3**	**OR**	Write $D = \{1, 2, 3, 6\}$ in set builder form. Also write $G = \{15x \mid x \in Z \wedge x \geq 1\}$ in tabular (roster) form.	**3**
2(ii)	If $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 4, 1, 7, 6\}$, find $A \cup B$ and $A \cap B$.	**3**	**OR**	If $U = \{1,2,3,4,5,6,7\}$, $A = \{2,4,6\}$, find A^c. Also verify $A \cup A^c = U$ and $A \cap A^c = \varnothing$.	**3**
2(iii)	If $A = \{1,2,3,4,5\}$ and $B = \{3,4,6,7,8\}$, find $A - B$ and $B - A$. Verify $A - B \neq B - A$.	**3**	**OR**	Define disjoint sets and overlapping sets. Give one example of each with a Venn diagram sketch.	**3**
2(iv)	If $X = \{a,b,c,d,e\}$, $Y = \{a,c,e\}$, $Z = \{g,h,i,j\}$, find $(X \cup Y) \cup Z$ using Venn diagram.	**3**	**OR**	If $P = \{0,1,2,3\}$, $Q = \{2,3,4,5,6\}$, $R = \{5,6,7,8,9\}$, find $P \cap (Q \cup R)$ using Venn diagram.	**3**
2(v)	Verify the associative property of intersection for $A = \{1,2,3,5,7,9\}$, $B = \{1,2,3,4,5,6\}$, $C = \{5,6,7,8\}$. Show $(A \cap B) \cap C = A \cap (B \cap C)$.	**3**	**OR**	Verify $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ for the same sets A, B, C.	**3**
2(vi)	In a class of 80 students, 40 like English, 34 like Mathematics and 9 like both. How many like either or both? How many like neither?	**3**	**OR**	Let $n(A) = 24$, $n(B) = 18$ and $n(A \cup B) = 31$. Find $n(A \cap B)$.	**3**
2(vii)	In a street with 50 houses, 25 have lawns, 32	**3**		Out of 70 people, 48 like tea and 40 like	**3**

	have a car porch and 15 have both. Find how many houses have neither.		**OR**	coffee and each person likes at least one drink. How many like both?	
2(viii)	Find x and y when $(x - 3y, 5x + 1) = (4, 6)$.	**3**	**OR**	Find the unknowns when $(2a - 4, 6) = (8, -b + 1)$.	**3**
2(ix)	Let $A = \{1,4,8\}$ and $B = \{1,0\}$. Find $A \times B$ and $B \times A$. How many elements are in each?	**3**	**OR**	If $L \times M = \{(0,2),(0,3),(0,4),(1,2),(1,3),(1,4)\}$, find sets L and M. Also find $M \times L$.	**3**
2(x)	Let $A = \{1,2,3\}$, $B = \{0,1,3\}$ and $R = \{(a,b) \mid a \in A, b \in B, a > b\}$. Write R in tabular form and show by an arrow diagram.	**3**	**OR**	Let $C = \{2,4,6\}$, $D = \{4,6,8,9,12\}$ and $R = \{(x,y) \mid x \in C, y \in D, x \text{ is a factor of } y\}$. Write R in tabular form and find Dom R and Range R.	**3**
2(xi)	Given $A = \{0,2,3\}$, $B = \{0,2,4,6,9,16\}$, $R = \{(x,y) \mid x \in A, y \in B, x^2 = y\}$. Find R, Dom R, Range R, R^{-1} and verify Dom R^{-1} = Range R.	**3**	**OR**	If $R = \{(2,0),(4,2),(6,4),(8,6),(10,8)\}$, write R in set builder form, find Dom R and Range R. Also write R^{-1} in tabular form.	**3**

SECTION C — Long Questions (4 × 5 = 20 Marks)

Attempt ALL 4 questions. Choose either the main question OR its alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	Verify both distributive laws through Venn diagram for $U = \{x : x \in W \wedge x \leq 12\}$, $A = \{x : x \in W \wedge x \leq 4\}$, $B = \{y : y \in N \wedge 3 \leq y \leq 7\}$, $C = \{z : z \in N \wedge 3 \leq z \leq 7\}$. (a) Verify $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (b) Verify $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	**5**	**OR**	Verify associative properties of union and intersection through Venn diagram for $X = \{2x \mid x \in N \wedge x < 20\}$, Y = Set of first 6 natural multiples of 3, $Z = \{6x \mid x \in W \wedge x < 20\}$. (a) Show $(X \cup Y) \cup Z = X \cup (Y \cup Z)$ (b) Show $(X \cap Y) \cap Z = X \cap (Y \cap Z)$	**5**
3(ii)	In a survey of 60 people, 25 watch channel A, 16 watch channel B, 13 watch channel C, 4 watch both A and B, 7 watch both B and C, 8 watch both A and C, and 3 watch all three. Using $n(A \cup B \cup C)$ find: (i) number watching at least one channel, (ii) number watching none.	**5**	**OR**	100 candidates appeared in an exam. 45 passed Math, 40 passed Science, 50 passed Health. 12 passed Math & Science, 15 passed Science & Health, 20 passed Health & Math, 5 passed all three. (i) Draw a Venn diagram. (ii) How many passed at least one? (iii) How many passed none?	**5**
3(iii)	Given $A = \{1,3,5\}$, $B = \{2,4\}$, $C = \{6,7\}$. (i) Find $A \times (B \cup C)$ (ii) Find $(A \times B) \cup (A \times C)$ (iii) Verify $A \times (B \cup C) = (A \times B) \cup (A \times C)$ (iv) Find the number of subsets of $A \times B$.	**5**	**OR**	Given $D = \{a,e,i\}$, $E = \{a,c\}$, $F = \{b,c\}$. (i) Find $D \times (E \cap F)$ (ii) Find $(D \times E) \cap (D \times F)$ (iii) Verify $D \times (E \cap F) = (D \times E) \cap (D \times F)$ (iv) Exhibit $D \times E$ and $E \times D$ by arrow diagram. Is $D \times E = E \times D$?	**5**
3(iv)	Let $S = \{1,2,4,8\}$ and $T = \{3^0,3^1,3^2\}$. Write the following in tabular form and state Dom and Range of each: (i) $R_1 = \{(x,y) \mid x \in S, y \in T, x = y\}$ (ii) $R_2 = \{(x,y) \mid y < x\}$ (iii) $R_3 = \{(x,y) \mid x+y \in E\}$ (iv) Find R_1^{-1}; verify Dom R_1^{-1} = Range R_1.	**5**	**OR**	Let $A = \{0,1,3\}$ and $B = \{1,2,3,5,7\}$. Write $R = \{(x,y) \mid x \in A, y \in B, y = 2x+1\}$ in tabular form. Find R^{-1} and show both R and R^{-1} by arrow diagrams. Verify Dom R^{-1} = Range R and Range R^{-1} = Dom R.	**5**

Invigilator's Signature: ____________________ Candidate's Signature: ____________________

Chapters Covered: Ch 3 — Sets and Relations

— End of Question Paper —